Advanced Nmap Lab – Network Security

Objective: By the end of this lab, you will explore advanced Nmap techniques, covering additional scanning methods for different network layers, evasion techniques, and deeper vulnerability assessments.

Tools Required:

- Nmap installed on your system.
- A local network or virtual machine to scan.

Question 1: Advanced Host Discovery

Perform an ICMP echo request, timestamp request, and address mask request for host discovery.

Command:

```
nmap -PE -PP -PM <target_network>
```

Questions:

- How does each of these options affect host discovery?
- How do results compare with a standard `-sn` scan?

Question 2: TCP Connect and FIN Scan

Use a TCP Connect scan and a FIN scan to compare results when scanning a target.

Commands:

```
nmap -sT <target_IP>
nmap -sF <target_IP>
```

Questions:

- What are the key differences in behavior between TCP Connect and FIN scans?
- How does the FIN scan bypass some firewall rules?

Question 3: Deep Service Version and Banner Grabbing

Perform an aggressive version scan along with manual banner grabbing to gather deeper insights.

Commands:

```
nmap -sV --version-all <target_IP>
nmap --script=banner <target_IP>
```

Questions:

- How does using `--version-all` improve accuracy?
- What additional information is gathered from banner grabbing?

Question 4: Advanced OS Detection

Use a combination of TCP/IP fingerprinting and NSE scripts to improve OS detection accuracy.

Commands:

```
nmap -O --osscan-guess <target_IP>
nmap --script=os-detection <target_IP>
```

Questions:

- How does `--osscan-guess` differ from a standard `-O` scan?
- What are the limitations of automated OS detection?

Question 5: Aggressive Scan with Firewall Detection

Perform an aggressive scan while detecting firewalls and analyzing traceroute details.

Commands:

```
nmap -A --script=firewalk <target_IP>
```

Questions:

- What additional information does the `firewalk` script provide?
- How can this scan help identify firewall rules?

Question 6: Advanced UDP Scanning

Perform a more detailed UDP scan by increasing probe intensity.

Commands:

```
nmap -sU --top-ports 20 --script=udp-proto-scanner <target_IP>
```

Questions:

- How does `--top-ports` improve efficiency in UDP scans?
- What protocols were identified using the `udp-proto-scanner` script?

Question 7: Full Network Sweep with OS and Service Detection

Scan an entire subnet with OS detection and service version scanning.

Commands:

```
nmap -p- -sV -O <network_range>
```

Questions:

- How does scanning all ports affect scan duration?
- What trends can be identified across multiple hosts?

Question 8: IDS/IPS Evasion Techniques

Use fragmentation and decoy scans to test evasion against IDS/IPS.

Commands:

```
nmap -f <target_IP>
nmap -D RND:10 <target_IP>
```

Questions:

- How do fragmented packets affect detection?
- Did the decoy scan successfully hide the actual scanner?

Question 9: Exploiting NSE for Vulnerability Scanning

Run an NSE script to check for known vulnerabilities.

Commands:

```
nmap --script=vuln <target_IP>
```

Questions:

- What vulnerabilities were identified?
- How can NSE scripting be extended for custom scans?

Question 10: Exporting and Parsing Scan Data

Save scan results in multiple formats for deeper analysis.

Commands:

```
nmap -oX scan_results.xml -oN scan_results.txt -oG scan_results.gnmap
<target_IP>
```

Questions:

- How do different output formats vary?
- How can XML results be integrated with security tools?

Extra Credit: Firewall Manipulation and Custom Payloads

1. **Try a NULL, XMAS, or ACK scan:**

```
nmap -sN <target_IP>
nmap -sX <target_IP>
nmap -sA <target_IP>
```

2. **Craft a custom payload for scanning:**

```
nmap --data-length 25 <target_IP>
```

Questions:

- How do different scan techniques affect firewall behavior?
- How can adding custom payloads help bypass signature-based detection?

This lab provides an in-depth exploration of Nmap features beyond basic scanning, offering insights into advanced techniques for security professionals.